ITUH USED 27-10-2020PM PAPER

Code No: 117HN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, October/November - 2020
SOFT COMPUTING
(Common to CSE, IT)

Time: 2 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
- - -

1.a) When will hill climbing searche fail? Do steepest ascent hill climbing always find solutions? How might some problems be overcome in search?
b) What are the constituents of constraint satisfaction problem? Write the formal definition of a constraint over a set of variables. [7+8]

2. Prove that A* heuristic search algorithm is optimal when applied in conjunction with a monotonic heuristic. State the conditions under which the algorithm is complete. [15]

3.a) Explain why a multilayered perceptron does not learn if the initial weights and biases are all zero.
b) With a neat sketch, explain the back propagation network with necessary formulas.[7+8]

4.a) Summarize the characteristics and applications of Artificial Neural Networks.
b) What are the basic models of ANN based on connection topology? Explain them in brief. [7+8]

5.a) What is unsupervised learning? Discuss the Kohonen's self-organizing maps in detail.
b) Discuss in detail about learning vector quantization. [7+8]

6.a) Draw and explain the working of Hamming network.
b) Explain the architecture of counter propagation networks, its training and working.[7+8]

7.a) Define fuzzy relation and max-min composition of fuzzy relations.
b) Discuss about various membership functions used in fuzzy systems. [7+8]

8.a) What do you mean by crossover and mutation? Explain different types of crossover and mutation techniques.
b) How traveling salesmen problem optimize using genetic algorithm? Explain. [7+8]

--ooOoo--